Tommaso Bonfanti

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Education

ETH Zurich – MSc in Quantum Engineering Sept. 2025 – exp. Jan. 2028

GPA: 5.88/6

Relevant coursework: Quantum Information Processing, Quantum Optics, Trapped-Ion Physics, Quantum Sensing and Metrology

Ongoing: AI in the Sciences and Engineering, Quantum Error Correction, FPGA Design

Politecnico di Milano – BSc in Mathematical Engineering Sept. 2022 – July 2025

Final Grade: 110/110, cum laude | GPA: 29.75/30

Relevant coursework: HPC, Operations Research, Probability Theory, PDEs, Numerical Methods

Honours: Sole recipient of the 'Dott. James Reid Noble' scholarship (€5,000); Best Freshmen Award, top of the Information Engineering cohort; Full merit-based tuition waiver for outstanding GPA (2023–2025)

Research Experience

QuanTech Project – ETH Zurich & PSI Sept. 2026 – Jan. 2027

Tunable microlens arrays for fast readout in neutral-atom quantum processors

Four-month project at the Nanophotonics Systems Laboratory (ETH) and the Experimental Quantum Engineering group (ETH–PSI). Team of four, selected by commission review of a self-written proposal.

Currently optimizing the thermo-optic microlens array design for integration into the EQE neutral-atom setup.

Research Assistant – ETH Zurich June 2026 – Sept. 2026

Trapped Ion Quantum Information Group *(Prof. Jonathan Home)*

- Extended a QuTiP-based spin-squeezing simulation module to support time-dependent Hamiltonians and inhomogeneous couplings, with Monte Carlo sampling of Brownian phase noise.
- Designed in KiCad, assembled and bench-tested a 4-layer PCB converting the EMCCD camera's LVDS readout signal to LVTTL at 40 MHz.
- Rebuilt the team's experiment codebase in Python, replacing compiled C++ firmware loads with JSON pulse sequences interpreted at runtime on a custom AMD Zynq UltraScale+ RFSoC.

Research Project – ETH Zurich Feb. 2026 – June 2026

Optical trapping behind a transparent membrane | Photonics Lab *(Prof. Lukas Novotny)*

- Simulated the vectorial focal field through a stratified medium to predict how an inserted glass slab broadens the focus and weakens the trap; measured trap frequencies to test the model.
- Designed a UHV-compatible assembly in SolidWorks to hold a graphene membrane between the focusing lens and the trapped particle.
- Built the experiment from an empty optical table: assembled and pumped down a new vacuum chamber, aligned the trapping path and forward QPD detection.

Additional Experience

Board Member – Quantum Engineering Commission, ETH Zurich Feb. 2026 – Present

Public Relations

- Co-organized the 2026 ETH Quantum Hackathon, with more than 175 participants over 3 days.
- Manage the commission's social media presence and external communications.

Private Tutor – Physics and Mathematics for high-school students Mar. 2023 – July 2025

Alice Bomboniere – E-commerce operations for a small retail business Nov. 2022 – Dec. 2025

Skills

Programming: Python (NumPy, SciPy, QuTiP, PyTorch), C, MATLAB, Git, Linux

Languages: Italian (native), English (IELTS 8.0 – C1), German (basic)

Interests

Competitive Football, Basketball, Piano, 3D printing. Regular blood donor (AVIS).